

Cuda invalid configuration error when using the statevector GPU simulator

<https://github.com/Qiskit/qiskit-aer/issues/1646>

ROW 10

Cuda invalid configuration error when using the statevector GPU simulator with a large number of qubits #1646

[New issue](#)[Closed](#)

HaoTy opened on Nov 6, 2022 · edited by HaoTy

Edits · ...

Informations

Qiskit Aer version:

```
qiskit          0.39.2
qiskit-aer       0.11.1
qiskit-aer-gpu   0.11.1
qiskit-ibmq-provider 0.19.2
qiskit-optimization 0.4.0
qiskit-terra     0.22.2
```



- Python version: 3.9.12
- Operating system: CentOS Stream 8
- GPU: A100-SXM4-80GB
- CUDA version: 11.7.0

What is the current behavior?

Cuda gives an invalid configuration error when using the statevector GPU simulator with a large number of qubits (more than 30 in my case).

```
qiskit.exceptions.QiskitError: 'Circuit execution failed: ERROR: [Experiment 0] for_each: failed to syn
```



Note that `AerPauliExpectation` is used and not until $n=36$ the memory becomes insufficient:

```
qiskit.exceptions.QiskitError: 'Circuit execution failed: ERROR: [Experiment 0] Insufficient memory to
```



Steps to reproduce the problem

```
import numpy as np
import networkx as nx
from qiskit_aer import AerSimulator
from qiskit.algorithms import QAOA
from qiskit.algorithms.optimizers import SPSA
from qiskit.utils import QuantumInstance
from qiskit.opflow import AerPauliExpectation
from qiskit_optimization.applications import Maxcut
from qiskit.utils import algorithm_globals

n = 32
p = 1
seed = 0

graph = nx.random_regular_graph(3, n, seed)
problem = Maxcut(graph).to_quadratic_program()
H, offset = problem.to_ising()

algorithm_globals.random_seed = seed
backend = AerSimulator(method="statevector", device="GPU")
quantum_instance = QuantumInstance(
    backend=backend,
    seed_simulator=seed,
    seed_transpiler=seed,
)
```



Assignees

[doichanj](#)

Labels

[bug](#)

Type

No type

Projects

No projects

Milestone

No milestone

Relationships

None yet

Development

No branches or pull requests

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```
algorithm_globals.random_seed = seed
backend = AerSimulator(method="statevector", device="GPU")
quantum_instance = QuantumInstance(
    backend=backend,
    seed_simulator=seed,
    seed_transpiler=seed,
)

algorithm = QAOA(
    SPSA(),
    reps=p,
    quantum_instance=quantum_instance,
    expectation=AerPauliExpectation(),
)
algorithm._check_operator_ansatz(H)
energy_evaluation= algorithm.get_energy_evaluation(H)
parameters = 2 * np.pi * np.random.default_rng(seed).random(2 * p)
energy_evaluation(parameters)
```

What is the expected behavior?

The simulator should be able to evaluate the QAOA energy like it does when $n=30$ or below.



HaoTy added **bug** on Nov 6, 2022



hhorii assigned [doichanj](#) on Nov 7, 2022



[doichanj](#) on Nov 7, 2022

Collaborator ...

This issue is caused because Aer tries allocating chunks in all free memory on GPU for hybrid parallelization. I will make fix to allocate chunks about 80% of free memory.



doichanj mentioned this on Nov 7, 2022

[Fix for GPU simulator, memory allocation #1650](#)



hhorii closed this as **completed** on Nov 15, 2022